ADVANCED JAVA PROGRAMMING LAB

(Course Code: 23UPCSC1L06)

A programming laboratory record submitted to Periyar University, Salem

In partial fulfilment of the requirements for the degree of

MASTER OF COMPUTER APPLICATIONS

By

GOPIKA D

[Reg. No: U24PG507CAP004]



DEPARTMENT OF COMPUTER SCIENCE

PERIYAR UNIVERSITY

State University - NAAC 'A++' Grade - NIRF Rank 94

State Public University Rank 40 - SDG Institutions Rank Band: 11-50

SALEM – 636 011

(OCTOBER - 2026)

CERTIFICATE

This is to certify that the Programming Laboratory entitled "**ADVANCED JAVA PROGRAMMING LAB (23UPCSC1L06)"** is a Bonafide record work done by Mr. /Ms._____ Register No: _____ as partial fulfilment of the requirements for the degree of Master of Computer Applications, in the Department of Computer Science, Periyar University, Salem, during the Academic Year 2026-2027.

Staff In-charge Head of the Department

Submitted for the practical examination held on ……..……

Internal Examiner External Examiner

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SOURCE CODE:

Student.java

```
import java.lang.*;
import java.io.*;

interface Exam
{
   void percent_cal();
}
class Student {
   String name;
   int  roll_no, mark1, mark2;
   Student(String n, int r, int m1,int m2) {
        name = n;  roll_no = r;  mark1 = m1;  mark2 = m2;
   }
   void display()
   {
         System.out.println("====" + "\n" + "Student Details" + "\n" + "====" + "\n");
         System.out.println("Name of Student: " + name);
         System.out.println("Roll No. of Student: " + roll_no);
         System.out.println("Marks of Subject 1: " + mark1);
         System.out.println("Marks of Subject 2: " + mark2);
    }

    public static void main(String[] args) {
    Result R = new Result("Mr.X", 12, 93,84);
          R.display();
          R.percent_cal();
     }
}
```

```
class Result extends Student implementExam {
      Result(String n, int r, int m1, int m2) {
          super(n, r, m1, m2);
       }

        public void percent_cal() {
           int total = (mark1 + mark2);
           float percent = (total * 100) /200;
           System.out.println("Percentage: " + percent + "%");
         }

         void display() {
            super.display();
          }
}
```

OUTPUT:

```
Student Details
====

Name of Student: Mr.X
Roll No. of Student: 12
Marks of Subject 1: 93
Marks of Subject 2: 84
Percentage: 88.0%
```

Source Code:

Exception.java

```
class exception {
   public static void main(String args[]) {
   try {
         int a = 30, b = 0;  int c = a / b;
         System.out.println("Result = " + c);
      } catch (ArithmeticException e) {
         System.out.println("Can't divide a number by 0");
      }
   try {
         int num = Integer.parseInt(This is a Number..!");
         System.out.println(num);
      } catch (NumberFormatException e) {
         System.out.println("Number format exception");
      }
    try {
         int a[] = new int[5];
         a[6] = 9;
      } catch (ArrayIndexOutOfBoundsException e) {
         System.out.println("Array Index is Out Of Bounds");
      }
    try {
         Object[] objArray = new Integer[10];
         objArray[0] = "String";
      } catch (ArrayStoreException e) {
         System.out.println("ArrayStoreException: " + e.getMessage());
      }
```

```
    try {
        String a = null;
        System.out.println(a.charAt(0));
    } catch (NullPointerException e) {
        System.out.println("NullPointerException: " + e.getMessage());
    }
  }
}
```

OUTPUT:

```
_e42e8491/bin exception
Can't divide a number by 0
Number format exception
Array Index is Out Of Bounds
ArrayStoreException: java.lang.String
NullPointerException: null
```

Source Code:

AccessModifier.java

```
class Employee{
   public String name;
   protected int employeeId;
   private double salary;

   public Employee(String name,int employeeId,double salary){
     this.name=name;       this.employeeId=employeeId;       this.salary=salary;
   }

   public void displayDetails(){
     System.out.println("Name:"+name);
     System.out.println("Employee ID:"+employeeId);
     System.out.println("Salary:$"+salary);
   }
   protected void calculateBonus(){
     double bonus=salary*0.1;
     System.out.println("Bonus:$"+bonus);
  }
  }
  class Manager extends Employee{
      private String department;

      public Manager(String name,int employeeId,double salary,String department){
            super(name,employeeId,salary);
           this.department=department;
    }
```

```
  public void displayDetails(){
    super.displayDetails();
    System.out.println("Department:"+department);
  }
}

class Staff extends Employee{
  private int workingHours;
  public Staff(String name,int employeeId,double salary,int workingHours){
     super(name,employeeId,salary);
     this.workingHours=workingHours;
  }
  public void displayDetails(){
    super.displayDetails();
    System.out.println("Working Hours:"+workingHours);
  }
}
public class AccessModifier{
public static void main(String[] args){
    Manager manager=new Manager("Mr.X",101,50000.0,"Sales");
    Staff staff=new Staff("Mr.Y",202,60000.0,50);
    System.out.println("Manager Details:");
    manager.displayDetails();
    manager.calculateBonus();

    System.out.println();
    System.out.println("Staff Details:");
   staff.displayDetails();
   staff.calculateBonus();
  }
}
```

OUTPUT:

```
Manager Details:
Name:Mr.X
Employee ID:101
Salary:$50000.0
Department:Sales
Bonus:$5000.0

Staff Details:
Name:Mr.Y
Employee ID:202
Salary:$60000.0
Working Hours:50
Bonus:$6000.0
```

Source Code:

MetricConversionApp.java

```
import javax.swing.*;
import java.awt.*;

public class MetricConversionApp extends JFrame {
    private JLabel cmLabel;
    private JTextField cmTextField;
    private JLabel inchLabel;
    private JTextField inchTextField;
    private JButton convertButton;

    public MetricConversionApp() {
      setTitle("Metric Conversion");
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      setSize(300, 150);
      setLayout(new FlowLayout());

      cmLabel = new JLabel("Centimeters:");
      cmTextField = new JTextField(10);
       inchLabel = new JLabel("Inches:");
       inchTextField = new JTextField(10);
       convertButton = new JButton("Convert");

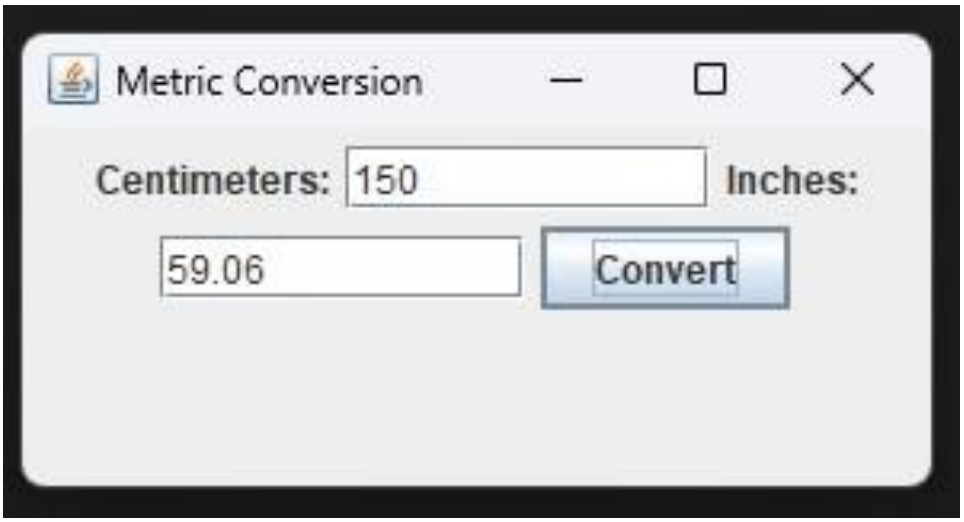
      convertButton.addActionListener(e -> {
         try {
            double centimeters = Double.parseDouble(cmTextField.getText());
            double inches = centimeters / 2.54;
            inchTextField.setText(String.format("%.2f", inches));
         } catch (NumberFormatException ex) {
```

```
                JOptionPane.showMessageDialog(null, "Invalid input. Please enter a valid number.");
            }
        });

        add(cmLabel);
        add(cmTextField);
        add(inchLabel);
        add(inchTextField);
        add(convertButton);
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            MetricConversionApp app = new MetricConversionApp();
            app.getRootPane().setDefaultButton(app.convertButton);
            app.setVisible(true);
        });
    }
}
```

OUTPUT:



Source Code:

Calc.java

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class calc extends JFrame implements ActionListener {
   private JTextField textField;
   private JButton[] buttons;
   private String input = "";
   private double num1, num2, result;
   private char operator;   public calc() {
   setTitle("Simple Calculator");   setSize(300, 400);
   setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
   setLayout(new BorderLayout());

   textField = new JTextField(10);
   textField.setFont(new Font("Arial", Font.PLAIN, 24));
   textField.setHorizontalAlignment(JTextField.RIGHT);
   textField.setEditable(false);
    JPanel panel = new JPanel(new GridLayout(4, 4));
    String[] buttonLabels = {"7", "8", "9", "+", "4", "5", "6", "-", "1", "2", "3", "*", "C", "0", "=", "/"};
    buttons = new JButton[buttonLabels.length];

  for (int i = 0; i < buttonLabels.length; i++) {
      buttons[i] = new JButton(buttonLabels[i]);
      buttons[i].addActionListener(this);
      panel.add(buttons[i]);
   }
```

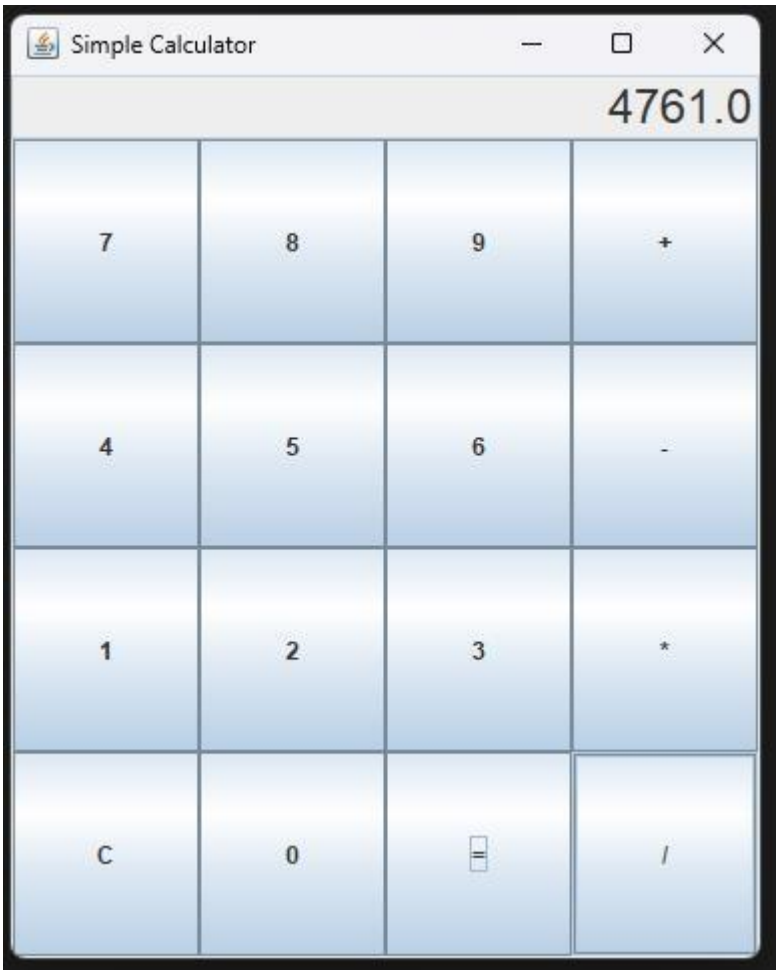
```
    add(textField, BorderLayout.NORTH);
    add(panel);
    setVisible(true);
 }

 public void actionPerformed(ActionEvent e) {
 String command = ((JButton) e.getSource()).getText();

 if (command.matches("[0-9]")) {
       input += command;
       textField.setText(input);
 }
else if (command.matches("[+\\-*/]")) {
        num1 = Double.parseDouble(input);
       operator = command.charAt(0);
       input = "";
 }
else if (command.equals("=")) {
        num2 = Double.parseDouble(input);
        switch (operator) {
            case '+':
            result = num1 + num2;
            break;
           case '-':
            result = num1 - num2;
            break;
            case '*':
            result = num1 * num2;
            break;
           case '/':
            result = num1 / num2;
```

```
        break;
      }
      textField.setText(String.valueOf(result));
      input = String.valueOf(result);
     }
     else if (command.equals("C")) {
         input = "";
         textField.setText("");
     }
  }
  public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> new calc());
  }
}
```

OUTPUT:



Source Code:

MyServer.java

```
import java.net.*;
import java.io.*;
class MyServer
{
   public static void main(String args[])throws Exception {
         ServerSocket ss=new ServerSocket(3333);
         Socket s=ss.accept();
         DataInputStream din=new DataInputStream(s.getInputStream());
         DataOutputStream dout=new DataOutputStream(s.getOutputStream());
         BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
         String str="",str2="";
         while(!str.equals("stop"))
         {
                  str=din.readUTF();
                  System.out.println("client says:"+str);
                  str2=br.readLine();
                  dout.writeUTF(str2);
                  dout.flush();
         }
         din.close();
         s.close();
         ss.close();
      }
}
```

MyClient.java

```
import java.net.*;
import java.io.*;
class MyClient {
	public static void main(String args[])throws Exception {
		Socket s=new Socket("localhost",3333);
		DataInputStream din=new DataInputStream(s.getInputStream());
		DataOutputStream dout=new DataOutputStream(s.getOutputStream());
		BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
		String str="",str2="";
		while(!str.equals("stop"))
		{
			str=br.readLine();
			 dout.writeUTF(str);
			dout.flush();
			str2=din.readUTF();
			System.out.println("Server says:"+str2);
		}
		dout.close();
		s.close();
	}
}
```

OUTPUT:

```
Client sr/bin/env /usr/lib/jvm/java-11-openjdk-amd64/bin/java -cp /home/assassinvijay73/.vscode-server/data/User/wo
rkspaceStorage/6a9f2770ff347152d04f61bf0b61955b/redhat.java/jdt_ws/htdocs_e42e8491/bin My
HELLO
Server says:HELLO
I  AM  CLIENT
Server says: I AM  SERVER
```

```
Server sr/bin/env /usr/lib/jvm/java-11-openjdk-amd64/bin/java -cp /home/assassinvijay73/.vscode-server/data/User/wo
rkspaceStorage/6a9f2770ff347152d04f61bf0b61955b/redhat.java/jdt_ws/htdocs_e42e8491/bin My
HELLO
client says:HELLO
 I AM  SERVER
client says:I  AM  CLIENT
```

Source Code:

Client.java

```
import java.rmi.Naming;

public class Client {
    public static void main(String[] args) {
        try {
            MyRemoteInterface remoteObject = (MyRemoteInterface) Naming.lookup
                                                ("rmi://localhost/MyRemoteObject");

            String result = remoteObject.sendMessage("Hello from RMI Client!");
            System.out.println("Server Response: " + result);        } catch (Exception e) {
            e.printStackTrace();
         }
     }
}
```

Server.java

```
import java.rmi.Naming;
import java.rmi.registry.LocateRegistry;

public class Server {
   public static void main(String[] args) {
        try {
            MyRemoteInterface remoteObject = new ServerImple();
            LocateRegistry.createRegistry(1099);
            Naming.rebind("MyRemoteObject", remoteObject);
            System.out.println("RMI Server is running...");
         } catch (Exception e) {
         e.printStackTrace();
```

```
            }
        }
}
```

MyRemoteInterface.java

```
import java.rmi.Remote;
import java.rmi.RemoteException;

public interface MyRemoteInterface extends Remote {
    String sayHello() throws RemoteException;
    String sendMessage(String message) throws RemoteException;
}
```

ServerImple.java

```
import java.rmi.RemoteException;
 import java.rmi.server.UnicastRemoteObject;

public class ServerImple extends UnicastRemoteObject implements MyRemoteInterface {
    public ServerImple() throws RemoteException {
    }

    @Override
    public String sayHello() throws RemoteException {
        return "Hello from RMI Server!";
    }

    @Override
    public String sendMessage(String message) throws RemoteException {
         return "Message received by the server: " + message;
    }
}
```

OUTPUT:

```
PS C:\Users\assas\Downloads\RMI (1)>  & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:50085' '-XX:+ShowCodeDetailsInExceptionMessages' '-
cp' 'C:\Users\assas\AppData\Roaming\Code\User\workspaceStorage\8164192c3e85e7ee91753177ba6e5827\redhat.java\jdt_ws\RMI (1)_ef11b9cf\bin' 'Server'
RMI Server is running...
```

```
PS C:\Users\assas\Downloads\RMI (1)>  & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:50099' '-XX:+ShowCodeDetailsInExceptionMessages' '-
cp' 'C:\Users\assas\AppData\Roaming\Code\User\workspaceStorage\8164192c3e85e7ee91753177ba6e5827\redhat.java\jdt_ws\RMI (1)_ef11b9cf\bin' 'Client'
Server Response: Message received by the server: Hello from RMI Client!
PS C:\Users\assas\Downloads\RMI (1)>
```

Source Code:

HelloServlet.java

```
package demo;
import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.*;
import javax.servlet.http.*;

@WebServlet("/HelloServlet")
public class HelloServlet extends HttpServlet {

    // Handle both GET and POST
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("username");
        if (name == null || name.isEmpty()) {
            name = "Guest";
        }
        out.println("<h2>Hello, " + name + "! Welcome to Servlet.</h2>");
    }

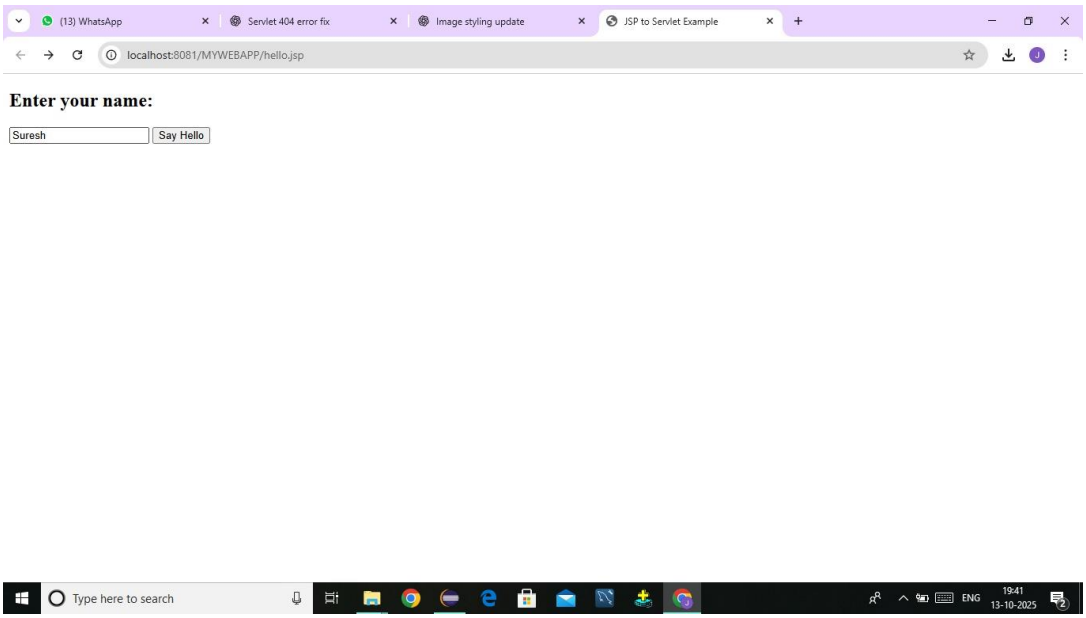
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
```

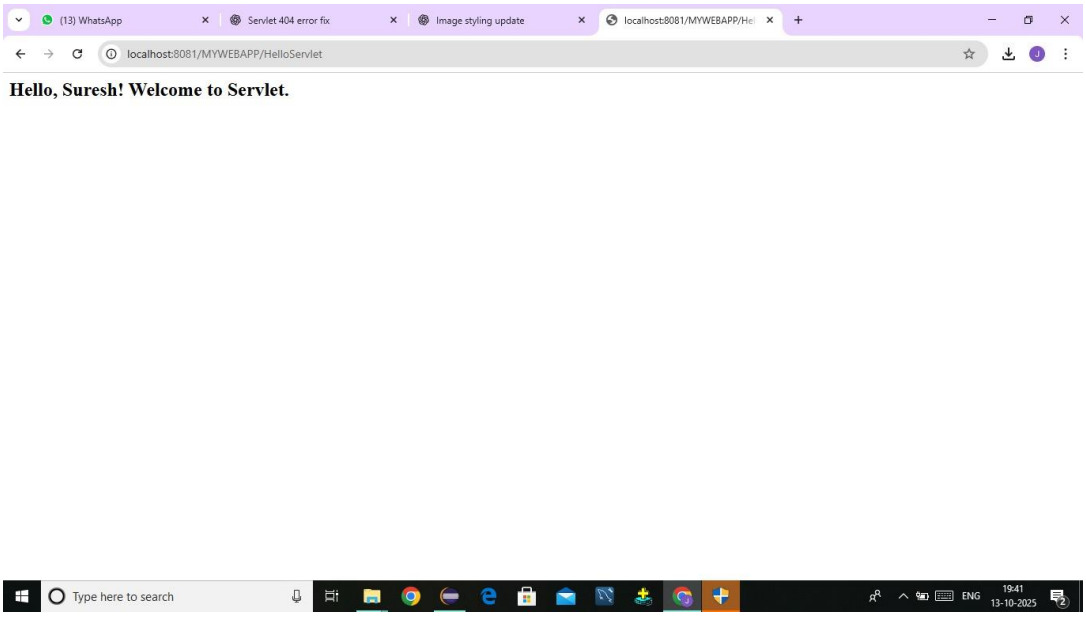
```
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
}
```

hello.jsp

```
<!DOCTYPE html>
<html>
<head>
<title>JSP to Servlet Example</title>
</head>
<body>
  <h2>Enter your name:</h2>
  <form action="HelloServlet" method="post">
    <input type="text" name="username" />
    <input type="submit" value="Say Hello" />
  </form>
</body>
</html>
```

OUTPUT:


(13) WhatsApp
Servlet 404 error fix
Image styling update
JSP to Servlet Example
localhost:8081/MYWEBAPP/hello.jsp
Enter your name:
Suresh
Say Hello
Type here to search
ENG
19:41
13-10-2025


(13) WhatsApp
Servlet 404 error fix
Image styling update
localhost:8081/MYWEBAPP/Hel
localhost:8081/MYWEBAPP/HelloServlet
Hello, Suresh! Welcome to Servlet.
Type here to search
ENG
19:41
13-10-2025

Source Code:

JavaApplication2.java

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class JavaApplication2 {
    public static void main(String[] args) {
        Connection c = null;
        Statement stmt = null;
        try {
          Class.forName("org.postgresql.Driver");
          c = DriverManager.getConnection("jdbc:postgresql://localhost:5432/emp", "postgres",
                                                                                    "Periyar");
          c.setAutoCommit(false);
          System.out.println("Open Database Connection Successfully");
          stmt = c.createStatement();
          ResultSet rs = stmt.executeQuery("SELECT * FROM emp_detail;");
          while (rs.next()) {
             int id = rs.getInt("eid");
             String name = rs.getString("ename");
             int age = rs.getInt("eage");
             String address = rs.getString("eaddress");
             float salary = rs.getFloat("esalary");
             System.out.println("ID = " + id);
             System.out.println("NAME = " + name);
             System.out.println("AGE = " + age);
             System.out.println("ADDRESS = " + address);
             System.out.println("SALARY = " + salary);
```

```
            System.out.println();
        }
        rs.close();
        stmt.close();
        c.close();
      } catch (Exception e) {
        System.err.println(e.getClass().getName() + ": " + e.getMessage());
        System.exit(0);
      }
      System.out.println("Operation Done Successfully");
    }
}
```

OUTPUT:

```
Open Database Connection Successfully
ID = 1
NAME = Mr.X
AGE = 30
ADDRESS = Salem, Chennai
SALARY = 50000.0

ID = 2
NAME = Mr.X
AGE = 25
ADDRESS = Salem, Chennai
SALARY = 55000.0

Operation Done Successfully
```

Source Code:

PersonBean.java

```
public class PersonBean {
    private String name;
    private int age;
    private String address;
    public String getName() {
          return name;
    }
    public void setName(String name) {
          this.name = name;
    }
    public int getAge() { return age; }
    public void setAge(int age) {
          this.age = age;
    }
    public String getAddress() {
          return address;
     }
     public void setAddress(String address) {
          this.address = address;
     }
}
```

JavaBean.java

```
public class JavaBean {
    public static void main(String[] args) {
        PersonBean person = new PersonBean();
        person.setName("Mr.V");
        person.setAge(21);
        person.setAddress("SALEM ");
        System.out.println("Name: " + person.getName());
        System.out.println("Age: " + person.getAge());
        System.out.println("Address: " + person.getAddress());
    }
}
```

OUTPUT:

```
PS C:\Users\assas\Desktop\New folder\java (1)> c:; cd 'c:\Users\assas\Desktop\New folder\java (1)'; & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:50360' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\assas\AppData\Roaming\Code\User\workspaceStorage\df099a7e5f0de3ee4429487099cad022\redhat.java\jdt_ws\java (1)_24fdad5\bin' 'JavaBean'
Name: Mr.V
Age: 21
Address: SALEM
```